

EDA Summary Report

Project 2: Exploratory Data Analysis (EDA), DecodeLabs Data Analytics Internship

Objective

Analyze the cleaned e-commerce order dataset (1,200 records) to understand patterns, trends, and distributions across key numeric variables, and identify any outliers requiring investigation.

Methodology

Descriptive statistics (count, mean, median, min, Q1, Q3, max, standard deviation) were calculated for four numeric fields: Quantity, UnitPrice, TotalPrice, and ItemsInCart. Outliers were identified using the IQR method (values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$). Correlation analysis (Pearson coefficient) was run across four variable pairs to check for meaningful relationships.

Descriptive Statistics

Metric	Quantity	UnitPrice	TotalPrice	ItemsInCart
Count	1200	1200	1200	1200
Mean	2.9458	356.4128	1053.968	5.485
Median	3	364.21	823.615	5
Min	1	11.39	11.39	1
Q1	2	186.0625	410.52	4
Q3	4	521.57	1578.475	7
Max	5	699.93	3456.4	10
Std Dev	1.4076	197.1771	819.8566	2.282

Correlation Analysis

Variable Pair	Pearson r	Interpretation
Quantity vs UnitPrice	0.0146	No relationship (independent)
Quantity vs TotalPrice	0.6153	Expected (TotalPrice derived from Quantity)
UnitPrice vs TotalPrice	0.7171	Expected (TotalPrice derived from UnitPrice)
ItemsInCart vs Quantity	0.6501	Genuine moderate-strong positive relationship

Key Findings

- Quantity, UnitPrice, and ItemsInCart are all symmetrically distributed, mean and median values are closely aligned across all three, with no outliers detected. These fields behave predictably with no data quality concerns.

- TotalPrice is right-skewed, mean (1,053.97) sits notably above the Median (823.62), indicating a small number of high-value orders are pulling the average upward.
- 8 statistical outliers identified in TotalPrice (0.67% of records, threshold >3,330.41). Investigation confirmed these are legitimate high-value transactions, not data errors, each combines maximum quantity (5 units) with premium-priced products (UnitPrice \$666–\$691). No correction was needed; $\text{TotalPrice} = \text{Quantity} \times \text{UnitPrice}$ validated with zero mismatches across all 1,200 records.
- ItemsInCart correlates moderately-strongly with Quantity ($r = 0.65$), customers who add more items to their cart tend to purchase in higher quantities per order. This is a genuine behavioral pattern, not a mathematical artifact (unlike the Quantity/UnitPrice $\rightarrow$ TotalPrice correlations, which are expected since TotalPrice is derived from those fields). Note: correlation does not imply causation, cart size may simply reflect a purchase decision already made, rather than driving it.
- Quantity and UnitPrice show no relationship ($r = 0.015$), confirms order size and product price point are independent in this dataset.

Observation / Business Relevance

The 8 high-value orders represent a small but consistent high-spend segment worth tracking separately (e.g., for loyalty targeting or bulk-order incentives). The cart-size/quantity relationship suggests cart behaviour could serve as an early signal for order value, useful for upsell or cart-abandonment strategies.